Native-Resolution Mask2Former for Solar Filament Instance Segmentation

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. Our solution is a single Mask2Former instance-segmentation network with a Swin-S backbone that trains and predicts at the native 2048×2048 resolution. Training mixes filament-centred crops with whole frames under exact dihedral augmentation, and a fixed post-processing rule turns the per-query masks into disjoint instances. On a validation split made by photograph and scored with the organisers' self-evaluation notebook, the model reaches a Panoptic Quality (PQ) of 0.4705 (95% bootstrap interval 0.450–0.491). It scores 0.41 on the public leaderboard, with one forward pass of 2.49 s per photograph on a T4 GPU.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

We treat the task as instance segmentation: each filament must be its own mask, and every annotation record, one annotator's labels of one photograph, is scored against the predictions for that photograph. Our submission is one model with one forward pass per photograph: no test-time augmentation and no ensemble. Section 2 describes the model, the data pipeline and the sampling strategy; Section 3 describes how we evaluated it, including a run of the organisers' self-evaluation notebook on our validation predictions.

2 METHODOLOGY

2.1 Segmentation algorithm

We use Mask2Former [3], an existing universal segmentation architecture, in its instance-segmentation form. A backbone, here Swin-S [8], extracts features at strides 4, 8, 16 and 32. A pixel decoder with multi-scale deformable attention [11] fuses them into per-pixel embeddings $\mathbf{E} \in \mathbb{R}^{C \times H/4 \times W/4}$. A Transformer decoder refines $N = 100$ learned queries; its cross-attention is *masked*, so each query attends only to the pixels its current mask covers. Each refined query $\mathbf{q}_i$ yields a class distribution over {filament, no object} and a mask

$$\hat{\mathbf{m}}_i = \sigma\left(\mathbf{q}_i^{\top} \mathbf{E}\right) \in [0,1]^{H/4 \times W/4}. \tag{1}$$

Training matches queries to ground-truth filaments one-to-one with the Hungarian algorithm [6] and minimises

$$\mathcal{L} = \lambda_{\text{cls}} \mathcal{L}_{\text{CE}} + \lambda_{\text{mask}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{dice}} \mathcal{L}_{\text{Dice}} \tag{2}$$

with $(\lambda_{\text{cls}}, \lambda_{\text{mask}}, \lambda_{\text{dice}}) = (2, 5, 5)$, the mask terms evaluated on sampled points and the no-object class weighted by 0.1, all as in [3]. We initialise from the COCO instance model [7], replace the class head by one foreground class and train with stochastic depth 0.3 (68.7 M parameters in total).

2.2 Data pipeline

Pre-processing. Photographs stay grey and at 2048×2048; the channel is repeated three times and normalised with ImageNet statistics. We never resize. The reason is the mask resolution: Mask2Former predicts masks at a quarter of its input. An ideal mask head on the stride-4 grid of a native input can reach a mean IoU of 0.914 with the annotated masks, but only 0.771 on the stride-8 grid that a 1024×1024 input would give (measured by downsampling the ground truth to each grid and back). Ground-truth polygons are rasterised at native resolution with pycocotools, the scorer's own convention.

Post-processing. The network outputs 100 candidates per photograph. The mask logits are bilinearly upsampled to 2048×2048 and thresholded at probability 0.5. Each candidate is scored as the product of its filament probability and its mean mask probability inside the mask. Candidates with a score of at least 0.85 are painted in descending score order; each keeps only the pixels that no higher-scoring candidate has claimed, and it is dropped if fewer than 100 pixels remain. The instances are therefore disjoint, as the scorer requires, and are written as RLE rows. The threshold and the minimum area were chosen once, by a sweep on validation over 19 thresholds (0.05 to 0.95), two minimum areas (100 and 300 pixels) and optional one-pixel erosion.

2.3 Sampling strategy and splits

The training file holds 1154 annotation records of 707 distinct photographs (8199 filaments); 296 photographs were annotated independently by two or three people. If two records of one photograph sat on opposite sides of a split, the model would be validated on a picture it had trained on, so we split by photograph with a fixed seed (2026): 601 photographs (974 records, 6874 filaments) for training and 106 photographs (180 records, 1325 filaments) for validation.

[1] www.esairlab.com/

Kaggle Competition, kaggle.com/competitions/filament-segmentation-2026 2026.

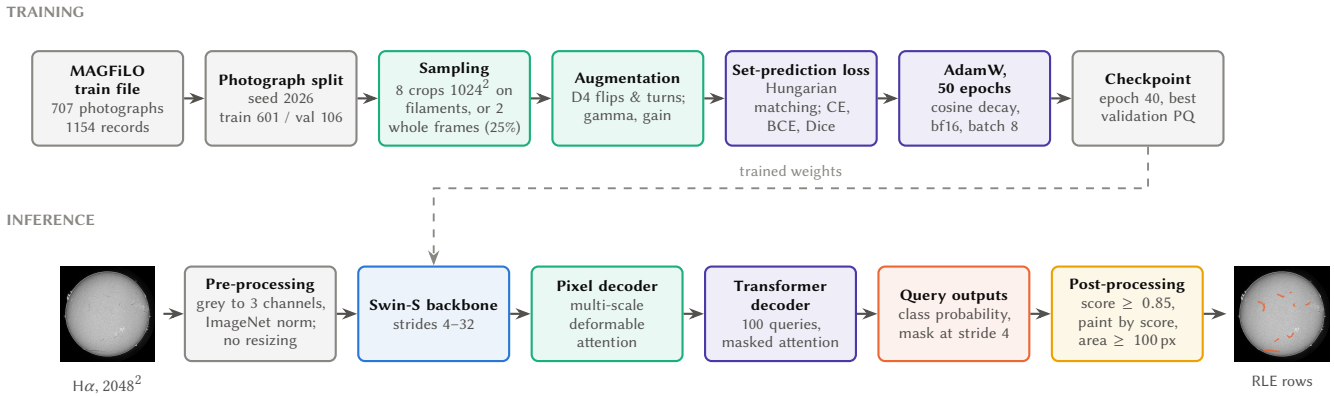


Figure 1: The end-to-end pipeline. Top: training, from the annotation file to the selected checkpoint. Bottom: inference, one forward pass per test photograph at native resolution in float32. Masks leave the network at stride 4 (512×512 per frame) and are upsampled to 2048×2048 before post-processing.

The 180 test photographs are held by the organisers. Every annotation record is a separate training sample, because the metric scores each record against one annotator.

Each optimiser step holds either eight 1024×1024 crops or two whole frames, so both kinds carry the same number of pixels; whole frames make up 25% of the steps. A crop is centred on a randomly chosen filament of its record, with jitter; 99% of training filaments are shorter than 516 pixels, so a crop almost always holds its filament whole. Whole frames show the model the full disk and the off-disk background at the scale it meets at inference. Augmentation applies the dihedral group D4 (flips and 90° turns, exact on the pixel grid), a gamma drawn log-uniformly from $[0.8, 1.25]$ and a gain from $[0.85, 1.15]$.

2.4 Training

We train with AdamW [9]: learning rate 10^{-4}, weight decay 0.05 (none on norms and biases), a 0.1× learning-rate multiplier on the backbone, gradient clipping at 0.01, 500 warm-up steps and cosine decay. We train for 50 epochs (12,150 steps) in bf16 on one H100 NVL GPU, at 0.47 s per step (1.6 hours) and 39.4 GiB peak memory. Every five epochs the run scores the validation split. We deploy the checkpoint of epoch 40, the best of these scores (Section 3). To make training affordable we replaced the pure-PyTorch deformable attention by a compiled CUDA kernel, batched the loss computation and moved photometric augmentation to the GPU; each change was accepted only after a numerical test against the original code.

3 EVALUATION

3.1 Evaluation strategy

A (ground truth, prediction) pair with IoU above 0.5 is a true positive; unmatched predictions are false positives and unmatched ground-truth filaments false negatives; the counts are pooled over all records and $\mathrm{PQ} = \sum_{\mathrm{TP}} \mathrm{IoU} / (|\mathrm{TP}| + \frac{1}{2}|\mathrm{FP}| + \frac{1}{2}|\mathrm{FN}|) = \mathrm{SQ} \times \mathrm{RQ}$ [5]. We ran the organisers' self-evaluation notebook verbatim on our validation predictions. It returns PQ 0.4705249989 and our own evaluator 0.4705249990, with identical counts, so all validation numbers

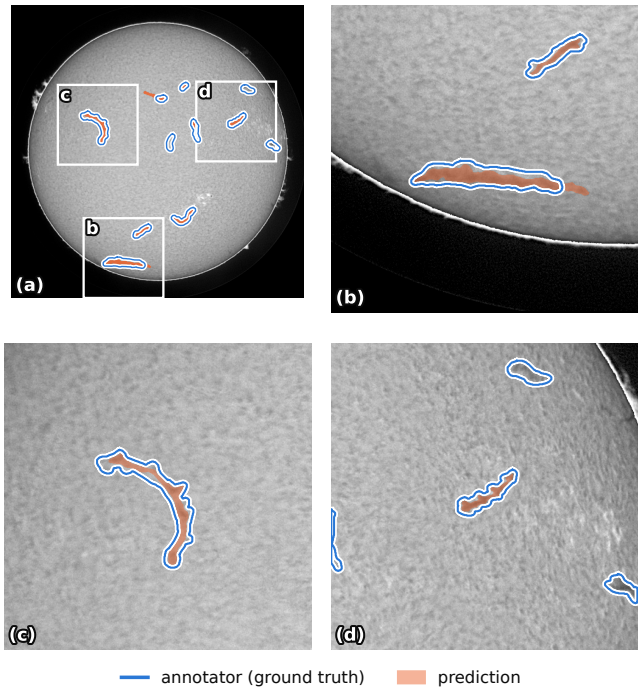


Figure 2: Segmentations on the validation photograph whose PQ (0.478) is closest to the median of the 34 single-annotator validation photographs with at least eight filaments (median 0.479); it is not a best case. (a) Full disk. (b)–(d) Native-resolution zooms on the three largest annotated filaments. Blue: the annotation, outlined 6 px (16 px on the disk) outside the mask so that the outline never covers it. Orange: our prediction. Display contrast is stretched.

here measure the leaderboard's quantity. Uncertainty comes from a bootstrap [4] that resamples the 106 validation photographs 10,000 times; a photograph's records share one set of predictions, so they

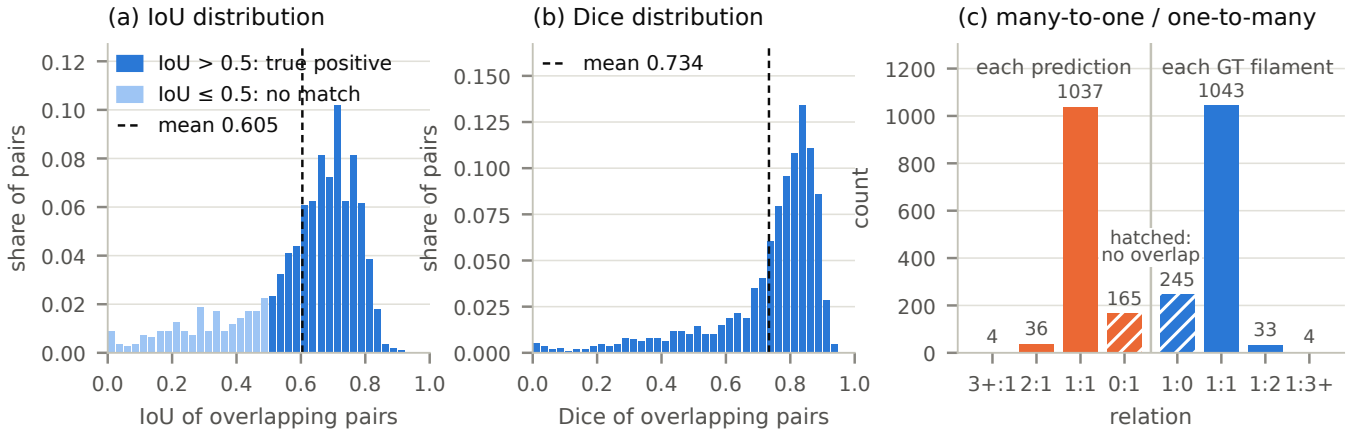


Figure 3: The self-evaluation notebook's measures on our validation predictions. (a) IoU of every overlapping (ground truth, prediction) pair; the darker bars pass the 0.5 threshold and count as true positives. (b) Dice of the same pairs. (c) Predictions touching each ground-truth filament (right) and ground-truth filaments touching each prediction (left); hatched bars have no overlap at all.

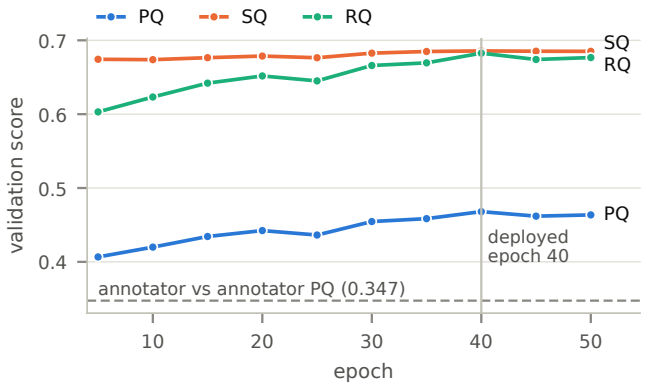


Figure 4: Validation PQ, SQ and RQ by epoch from the evaluation inside training (bf16 inference) at the deployed rule. Dashed: annotator-versus-annotator PQ.

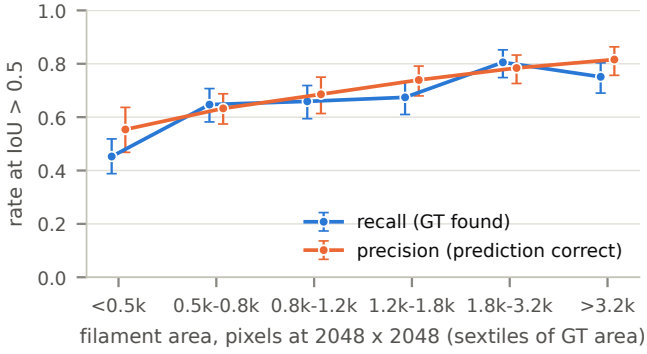


Figure 5: Recall (ground-truth filaments found at IoU > 0.5) and precision (predictions that match) by filament area, in sextiles of ground-truth area (221 filaments per bin). Bars: Wilson 95% intervals.

Table 1: Validation results (106 photographs, 180 records, host notebook). Brackets: 95% bootstrap intervals over photographs.

	PQ	SQ	RQ
Our model, epoch 40	0.4705 [0.450, 0.491]	0.685 [0.680, 0.691]	0.686 [0.657, 0.715]
TP / FP / FN	881 / 361 / 444		
Annotator vs annotator	0.347 [0.314, 0.383]	0.636	0.546
Public leaderboard (test)	0.41	–	–

are resampled together. Rates carry Wilson 95% intervals [10]. To read the scores we also scored each annotator against every other annotator of the same validation photograph under the same rule.

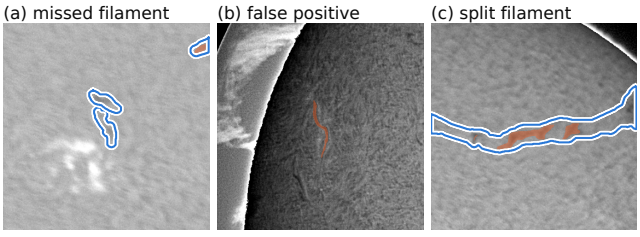


Figure 6: One example of each error, chosen by a fixed rule over single-annotator validation records: (a) the median-area filament that no prediction touches; (b) the median-area prediction that touches no filament; (c) the largest filament touched by two or more predictions. Colours as in Figure 2.

3.2 Results

Table 1 summarises the validation scores. Our model agrees with a held-out annotator more closely than two annotators agree with each other, and the two intervals do not overlap. A model trained on many annotators learns their average, which can sit closer to each of them than they sit to each other, so this is a reference for label variability rather than a claim of super-human skill. The public leaderboard shows 0.41. It is scored on a different set of photographs and annotators and shown to two decimals, so it is not directly comparable with the validation interval.

Figure 3 shows the distributions the notebook reports. Overlapping pairs have a mean IoU of 0.605 and a mean Dice of 0.734. Of the ground-truth filaments, 78.7% touch exactly one prediction, 2.8% touch two or more (fragmentation) and 18.5% touch none. Of the predictions, 83.5% touch exactly one filament, 3.2% touch two or more (merging) and 13.3% touch none.

Figure 4 shows the scores during training. PQ peaks at epoch 40 and then drifts down while SQ stays flat, so the late decline comes from recognition, not from mask quality. Because epoch 40 was selected on the same validation split, its validation score is slightly optimistic.

3.3 Strengths, weaknesses and improvements

Strengths. Filaments are rarely split or merged (2.8% and 3.2%): each query proposes one whole filament, and painting by score keeps the instances disjoint without a separate merging step. Masks are accurate once a filament is found (SQ 0.685). The whole test set runs in about 7 minutes on one T4 GPU (2.49 s per photograph, 2.7 GiB of memory).

Weaknesses. The model misses filaments far more often than it invents them: 444 false negatives against 361 false positives. Figure 5 shows where: recall is 0.45 [0.39, 0.52] for the smallest sixth of filaments (under 471 pixels) against 0.75 [0.69, 0.80] for the largest. Figure 6(a) shows a typical miss, a small faint structure (644 pixels). Some false positives are dark structures that the annotator did not mark (Figure 6(b)), and long filaments are occasionally covered by several predictions (Figure 6(c)).

Improvements. Three changes address these weaknesses directly. First, train on all 1154 records once the recipe is fixed, adding 18% more annotations. Second, recover small filaments with a verification stage that examines candidates scoring below the 0.85 threshold. Third, use the 296 multiply-annotated photographs to build consensus targets, which should reduce the noise in the training signal.

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